# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

**Department of Artificial Intelligence and Machine Learning**

## ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

| | | | |
|---|---|---|---|
| **Course Code:** | ITA05 | **Course Title:** | **Computer Vision** |
| **CO Assessed:** | CO2 | **Assessment:** | Assessment 2: Scenario-Based Questions |
| **Weightage:** | 35% of CIA | **Total Marks:** | (5 Questions × 10 Marks) |
| **CO2 Statement:** | ***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.*** | | |
| **Student Name:** | Thalamati Aasritha | **Reg. No.:** | 192311008 |
| **Section / Batch:** | CSE | **Date:** | 28/10/2026 |

**Code of Conduct**

*I* ~~Thalamati Aasritha / 192311008~~ *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

**QUESTIONS: 5** **Total Marks: 50**

## Annexure A

## Scenario-Based Qns

### Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

### Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

### Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

### Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

### Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

**Rubrics for Calculation**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Technical Content Accuracy** | 30 | Highly accurate, indepth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| **Problem Identification** | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| **Use of Tools** | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| **Communication** | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| **Professionalism** | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |

# ASSESSMENT 2 — SOLUTIONS

**Course:** Computer Vision **Course Code:** ITA05 **CO:** CO2

**Student:** Thalamati Aasritha **Register No.:** 192311008 **Batch:** CSE

The solutions below address the five scenario-based questions in Annexure A. Numerical values are illustrative examples because no actual input image or pixel dataset was supplied in the assessment.

## Q1. Image Enhancement Using Grey-Level Transformations

A dark X-ray can be enhanced by applying a grey-level transformation that maps low input intensities to brighter output intensities. A simple and suitable choice is the linear transformation **s = 1.5r + 20**, where *r* is the original 8-bit pixel intensity and *s* is the transformed intensity. Values above 255 are clipped to 255.

### Worked example

| Original intensity r | Calculation | Output s |
|---|---|---|
| 20 | 1.5(20) + 20 | 50 |
| 60 | 1.5(60) + 20 | 110 |
| 100 | 1.5(100) + 20 | 170 |
| 150 | 1.5(150) + 20 | 245 |
| 180 | 1.5(180) + 20 = 290 | 255 (clipped) |

**Explanation:** The offset raises dark pixel values, while the multiplier expands intensity differences and increases contrast. This can make faint anatomical structures easier to distinguish. In practice, the transformation should be adjusted carefully so bright regions do not saturate and important X-ray detail is not lost.

Conclusion: A linear brightness-and-contrast transformation improves visibility of structures in underexposed X-rays, provided clipping is limited.

## Q2. Histogram Processing and Spatial Filtering

Histogram equalization redistributes image intensity values using the cumulative histogram. It spreads commonly occurring grey levels across a wider range, improving global contrast when a night-time surveillance image looks flat or dim.

### Noise reduction

A 3 × 3 median filter is suitable for reducing impulse-like noise while retaining many edges. For each pixel, the nine values in its local neighbourhood are sorted and the middle value replaces the centre pixel. Unlike a simple averaging filter, the median filter is less affected by isolated extreme values.

### Comparison

| Technique | Main operation | Expected effect | Limitation |
|---|---|---|---|
| Histogram equalization | Remaps intensities using cumulative histogram | Improves overall contrast and visibility | May amplify noise or over-enhance some regions |
| 3 × 3 median filter | Replaces each pixel with neighbourhood median | Suppresses speckle/impulse noise and keeps edges relatively clear | May remove very fine details if applied repeatedly |

Recommended workflow: apply a mild median filter first to reduce noise, then use histogram equalization or adaptive histogram equalization to improve contrast. Inspect the result to ensure faces, clothing, and object boundaries remain visible.

Conclusion: Histogram equalization improves contrast; spatial filtering reduces noise. Used together and conservatively, they make surveillance images easier to interpret.

## Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

Frequency-domain filtering transforms an image from the spatial domain into its frequency representation, multiplies it by a filter function, and then applies the inverse transform to obtain the processed image.

### Low-pass filtering

A low-pass filter retains low spatial frequencies and attenuates high frequencies. A Gaussian low-pass filter is a practical choice because it reduces rapid intensity changes and high-frequency noise smoothly. It can make large land regions more uniform, although excessive smoothing can blur narrow roads, coastlines, and small field boundaries.

### High-pass filtering

A high-pass filter suppresses slowly varying background components and retains rapid intensity changes. These changes often correspond to edges such as coastlines, field boundaries, and roads. A high-pass result can be used for edge emphasis or added back to the original image for sharpening.

| Feature | Low-pass filter | High-pass filter |
|---|---|---|
| Frequencies retained | Low frequencies | High frequencies |
| Primary purpose | Smooth image; reduce fine noise | Emphasize edges and fine detail |
| Typical risk | Blurred boundaries and loss of detail | Noise amplification and artificial halos |
| Use in satellite images | Reduce small-scale noise before interpretation | Highlight boundaries between geographic regions |

Conclusion: Low-pass filtering supports noise reduction and broad-region interpretation, while high-pass filtering enhances boundaries. A balanced combination is preferable to applying either filter too strongly.

## Q4. Image Segmentation Using Edge Detection and Thresholding

For metal-surface inspection, the goal is to locate crack boundaries and isolate suspicious dark or bright regions from the background. A practical pipeline is to reduce noise, detect edges, threshold the defect response, and inspect the resulting binary mask.

### Method

**1. Pre-processing:** Convert the image to grayscale and apply a small Gaussian blur to suppress sensor noise. **2. Edge detection:** Use the Canny operator, which calculates image gradients, applies non-maximum suppression, and uses two thresholds with edge tracking by hysteresis. It is useful for identifying thin crack boundaries. **3. Thresholding:** Apply a global threshold when illumination is fairly uniform, or adaptive thresholding when brightness varies across the component. Pixels meeting the defect criterion are assigned 1 (white) and the remaining background pixels 0 (black). **4. Cleanup:** Use small morphological operations to remove isolated specks or close tiny gaps, while checking that real narrow cracks are not erased.

### Binary representation

| Pixel classification | Binary value | Meaning |
|---|---|---|
| Pixel identified as defect | 1 (white) | Candidate crack / defective region |
| Pixel classified as background | 0 (black) | Non-defect region |

How it helps: Canny detection highlights possible crack contours, while thresholding converts the image into a mask that can be measured and compared against inspection limits. The mask can support estimates of defect area, length, and location. Threshold values should be calibrated against representative good and defective components.

Conclusion: Edge detection locates candidate boundaries; thresholding creates a binary defect mask for consistent automated inspection.

## Q5. Region-Based Segmentation and Boundary Detection

Agricultural aerial images often contain adjoining regions with similar brightness, so segmentation should consider both pixel similarity and spatial connectivity. Region growing is a suitable region-based method when representative seed points can be selected for crop fields, water, and surrounding land.

### Procedure

**1. Pre-process:** Smooth minor noise while preserving large boundaries. **2. Select seeds:** Choose representative pixels inside known crop, water, or land areas. **3. Region growing:** Examine neighbouring pixels and add a pixel when its intensity, colour, or texture is sufficiently similar to the region model. Stop growth when the similarity criterion is not met. **4. Boundary detection:** Identify transitions between neighbouring region labels. **5. Edge linking:** Connect short gaps between compatible edge fragments using proximity, direction, and local continuity. Review linked edges to avoid joining unrelated features.

### Illustrative region labels

| | | | | | |
|---|---|---|---|---|---|
| C | C | C | L | L | L |
| C | C | C | L | W | W |
| C | C | L | L | W | W |
| C | L | L | W | W | W |
| L | L | L | W | W | W |

Legend: C = crop field; L = surrounding land; W = water body. This grid is a schematic example, not a segmentation result from a supplied aerial photograph.

How it helps: Region growing groups connected pixels into coherent land-cover areas. Edge linking repairs small discontinuities, and boundary detection outlines each region for mapping or area estimation. Accuracy depends on seed selection, similarity thresholds, image resolution, and the quality of the input image.

Conclusion: Combining region-based segmentation with edge linking produces more complete boundaries and supports clearer identification of crop fields, water bodies, and surrounding land.

## Overall Summary

Grey-level transformations improve brightness and contrast; histogram equalization improves contrast distribution; spatial and low-pass filters reduce noise; high-pass filtering and Canny detection emphasize boundaries; thresholding creates binary masks; and region growing groups similar neighbouring pixels. The method should be selected according to the image conditions and validated against the original image to avoid losing important detail.